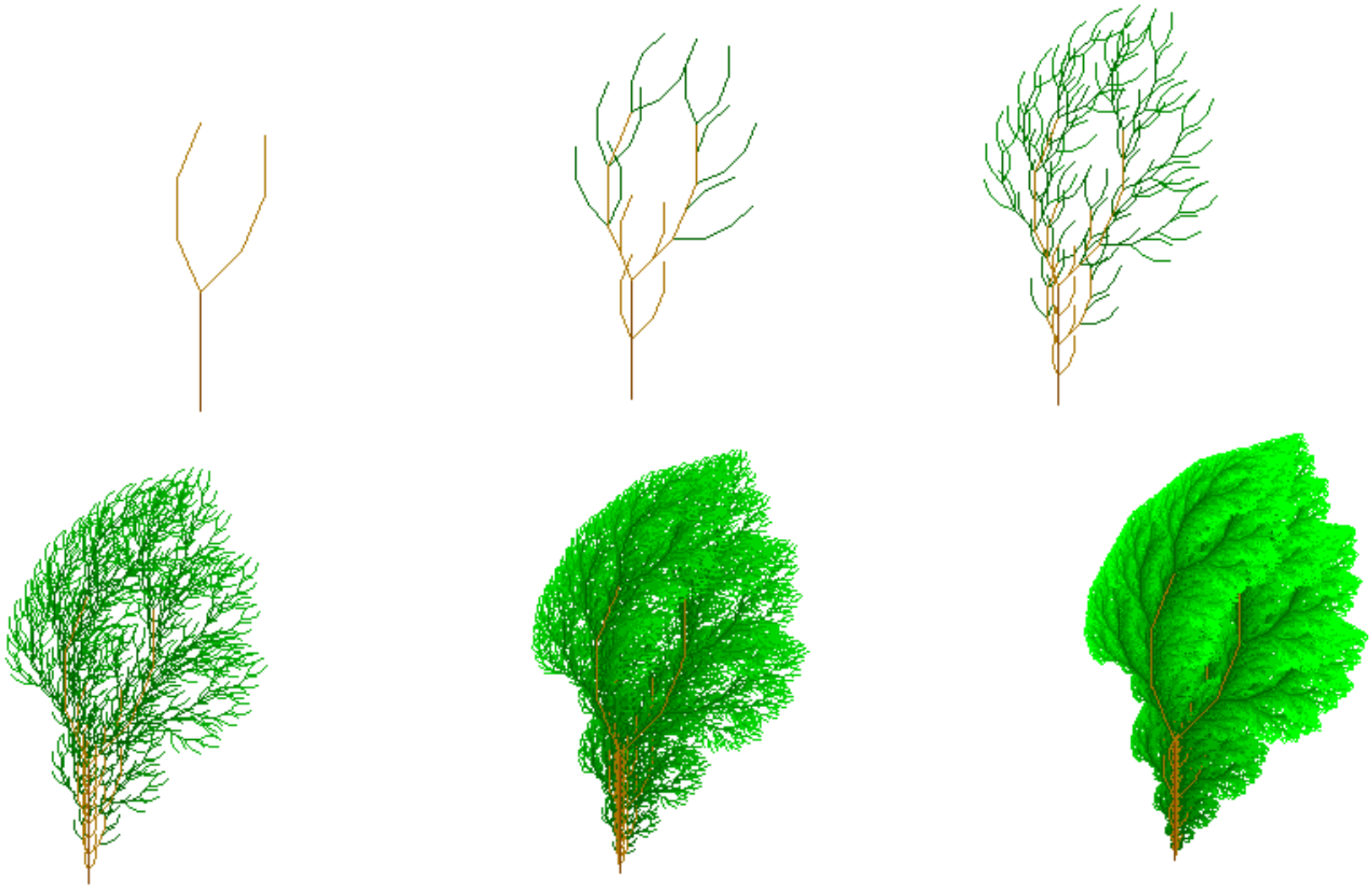
Representation and Modeling of 3D Objects (P2)

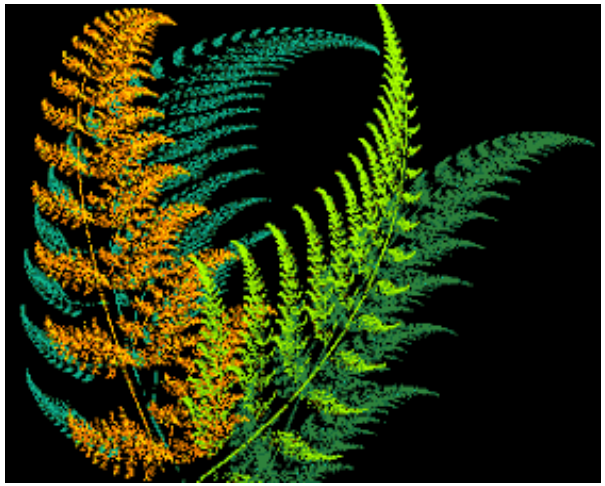
Contents

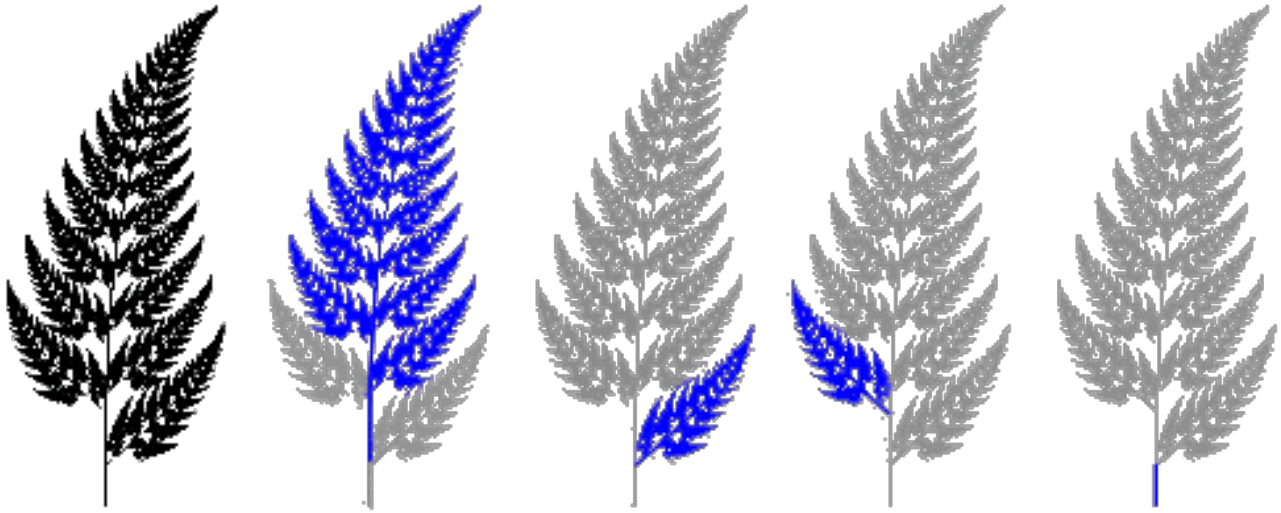
- Implicit functions
- Polygonal representation
- Bicubic parametric patches
- Constructive solid geometry
- Space subdivision techniques
- **Procedural modeling**
- Artificial intelligence

Fractals



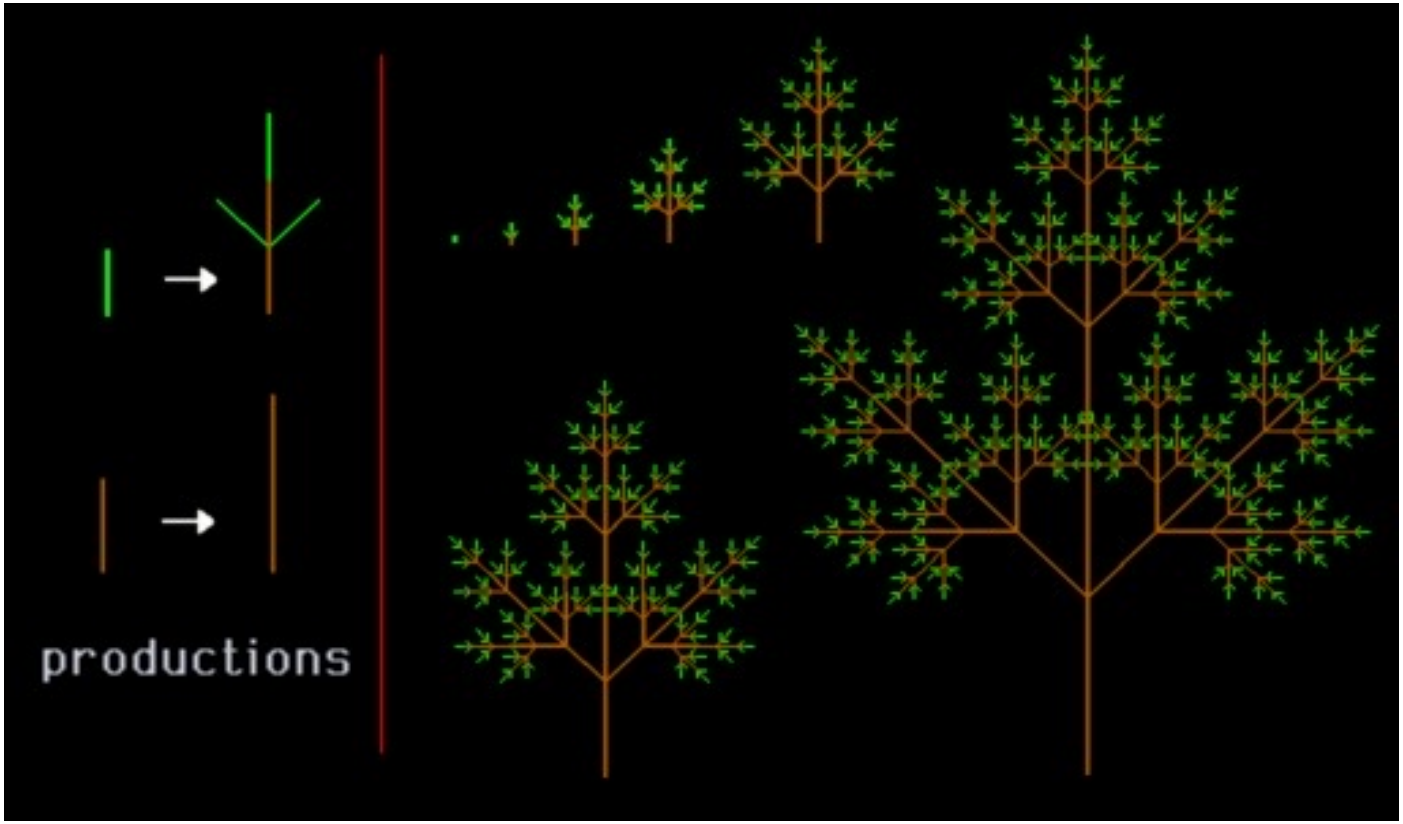
Fractals





Procedural modeling

- Lindenmayer-System or L-System
- Prusinkiewicz, P., Lindenmayer, A., Hanan, J.: "Development Models of Herbaceous Plants for Computer Imagery Purposes", SIGGRAPH 88, pp. 141-150
- Examples from Prusinkiewicz, Hammel, Mech. http://www.cpsc.ucalgary.ca/projects

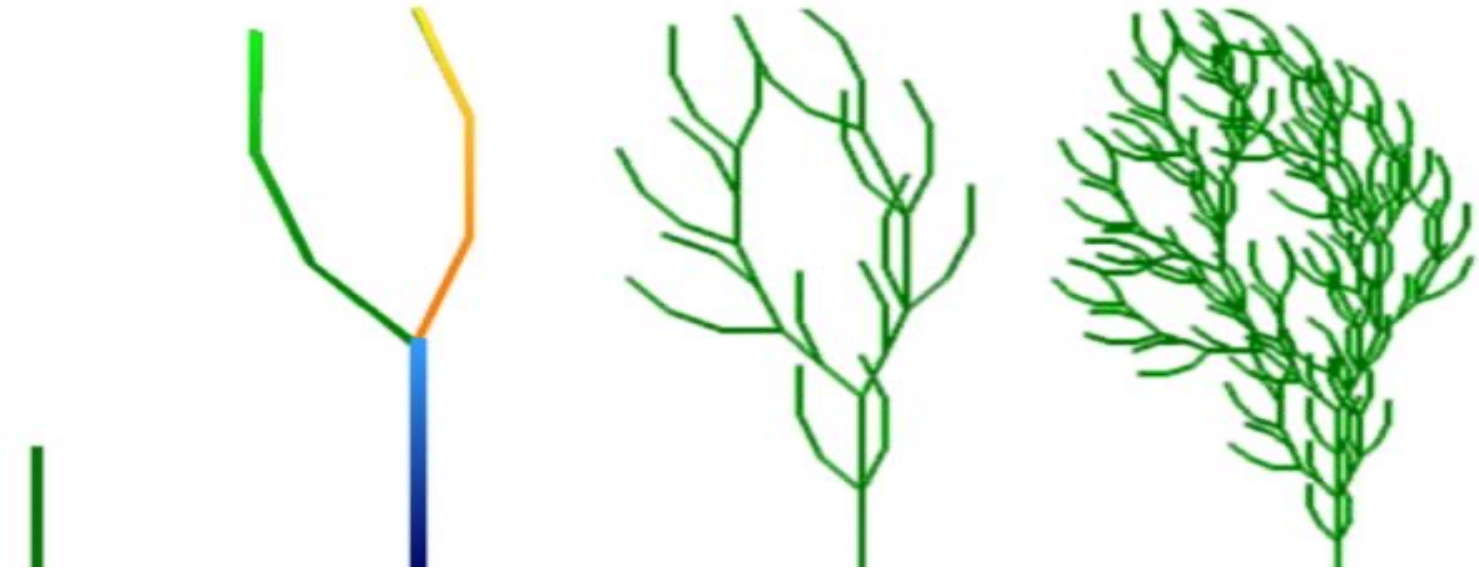


Lindenmayer-System or L-System

- Production rules:

  W -> F

  F -> FF+[+F-F-F] -[-F+F+F]



Lindenmayer-System or L-System





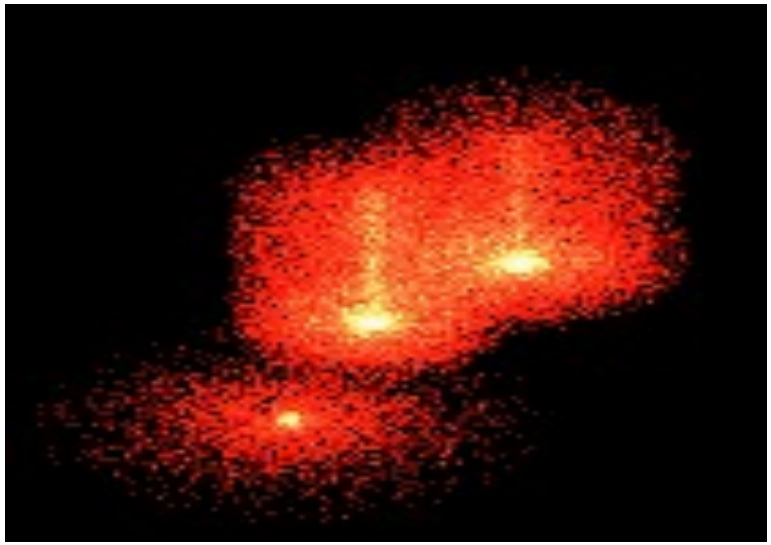
Lindenmayer-System or L-System





Procedural modeling

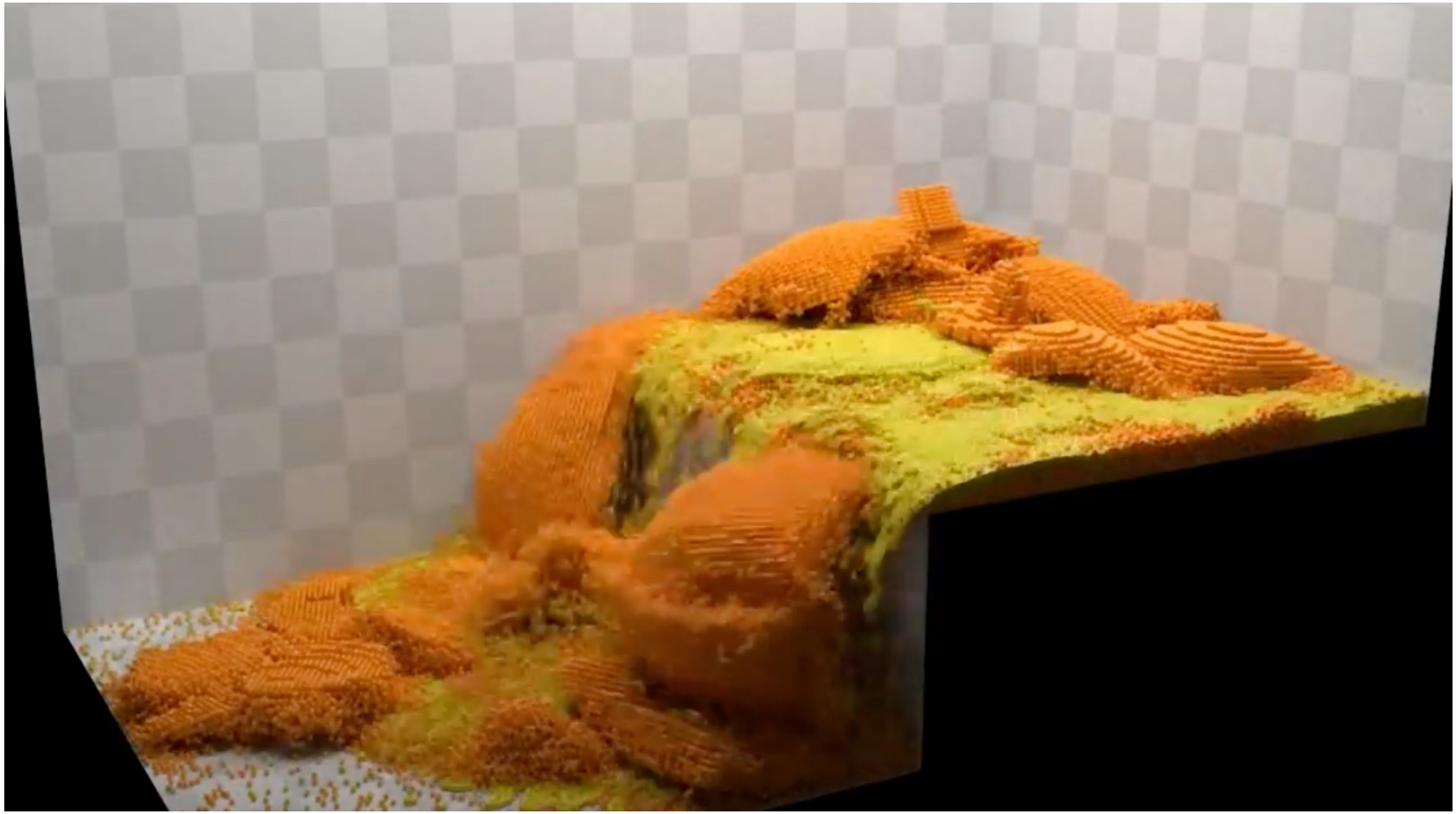
- Particle based systems
  - The term particle system is loosely defined in computer graphics. It has been used to describe modeling techniques, rendering techniques, and even types of animation. In fact, the definition of a particle system seems to depend on the application that it is being used for.
- Describe natural or irregular shaped objects which are flowing or fluid-like, particular objects which are changing over time
- Model objects like grass, waterfalls, fireworks, cloth, etc.



Particle based systems

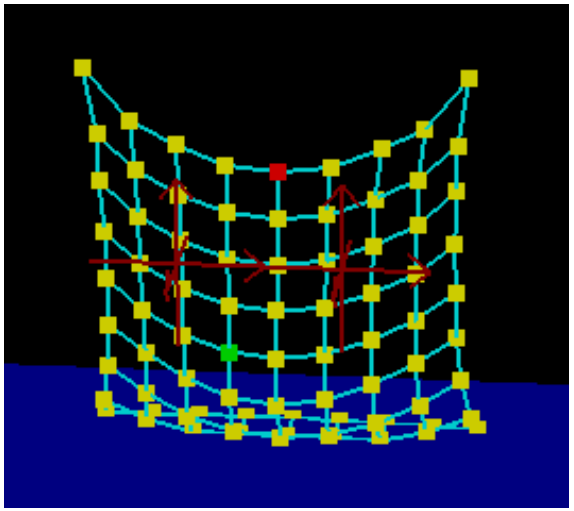
- Collection of particles - a particle system is composed of one or more individual particles.
- Stochastically defined attributes : position, velocity (speed and direction), color, lifetime, age, shape, size, transparency
- Particle Life Cycle
  1. Generation
     - particles in the system are generated randomly within a predetermined location of the fuzzy object
  2. Particle Dynamics
     -the attributes of each of the particles may vary over time. (particle position is going to be dependent on previous particle position and velocity as well as time)
  3. Extinction
     - each particle has two attributes dealing with length of existence: age and lifetime.
  4. Rendering
     - when rendering this system of thousands of particles, some assumptions have to be made to simplify the process.

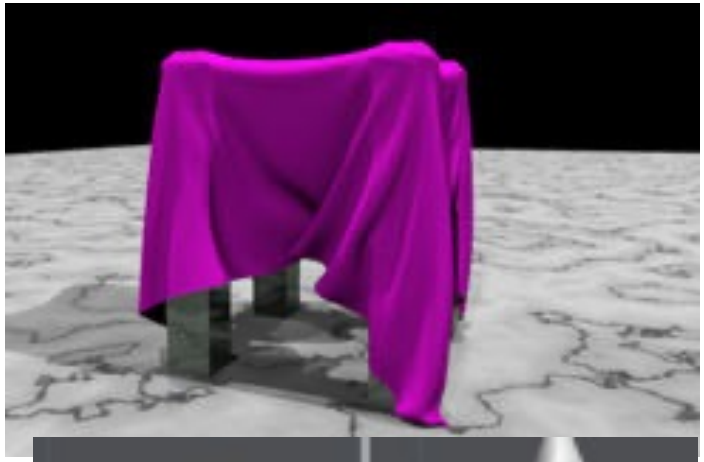
Particle based modeling



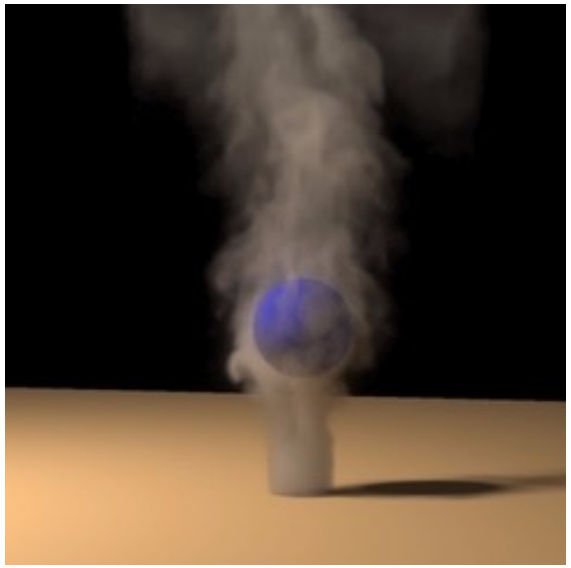
Advanced Molecular & Particle Physics Simulations,
https://www.youtube.com/watch?v=x8Fo2slT2WA

Particle based modeling





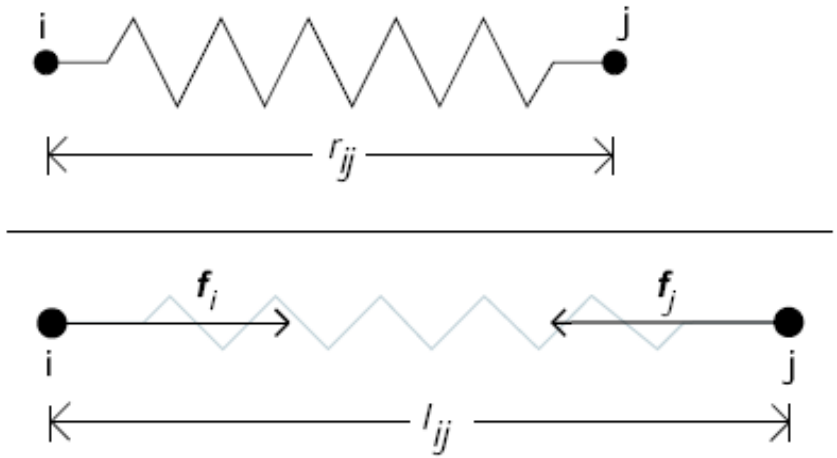




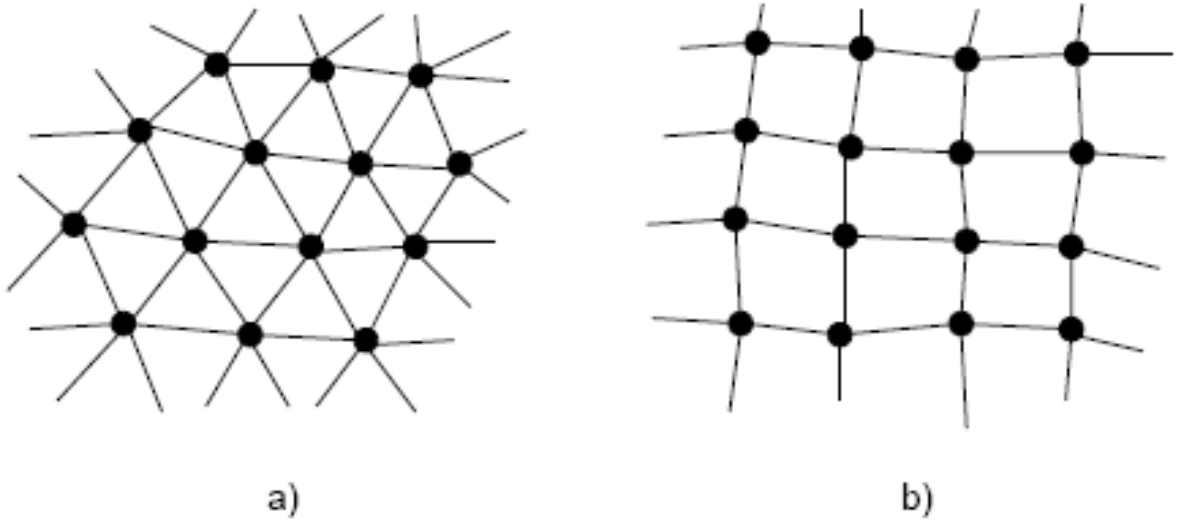


Mass-spring particle model

- Connected particle network
- The link between two nodes **i** and **j** is an arc
- $\Delta x = k\ x$



Mass-spring particle model



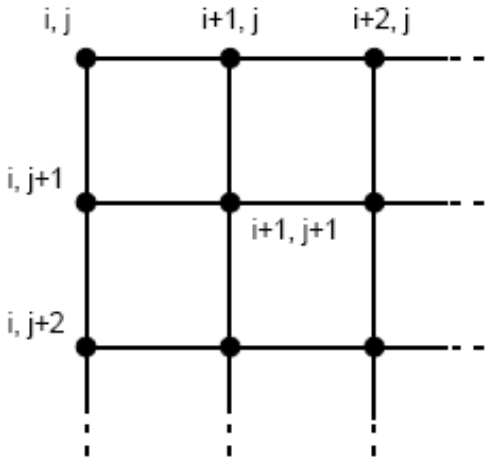
Particle network a) triangular topology b) rectangular topology.

Simulation:

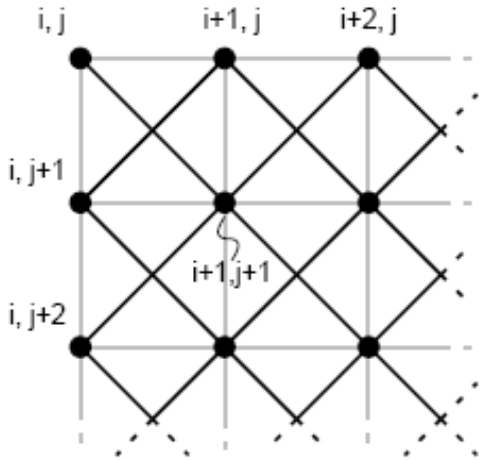
1. Force computation
2. Collision detection
3. Graphics rendering

Mass-spring particle model

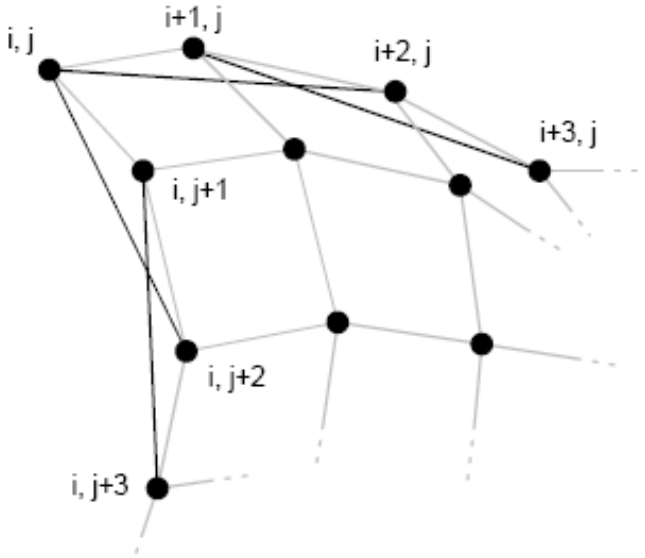
- Links between nodes (arcs):
  - Structural/Tensile springs (Structurale)
  - Shear springs (Intindere)
  - Bend springs (Indoire)



a) Structural/Tensile springs



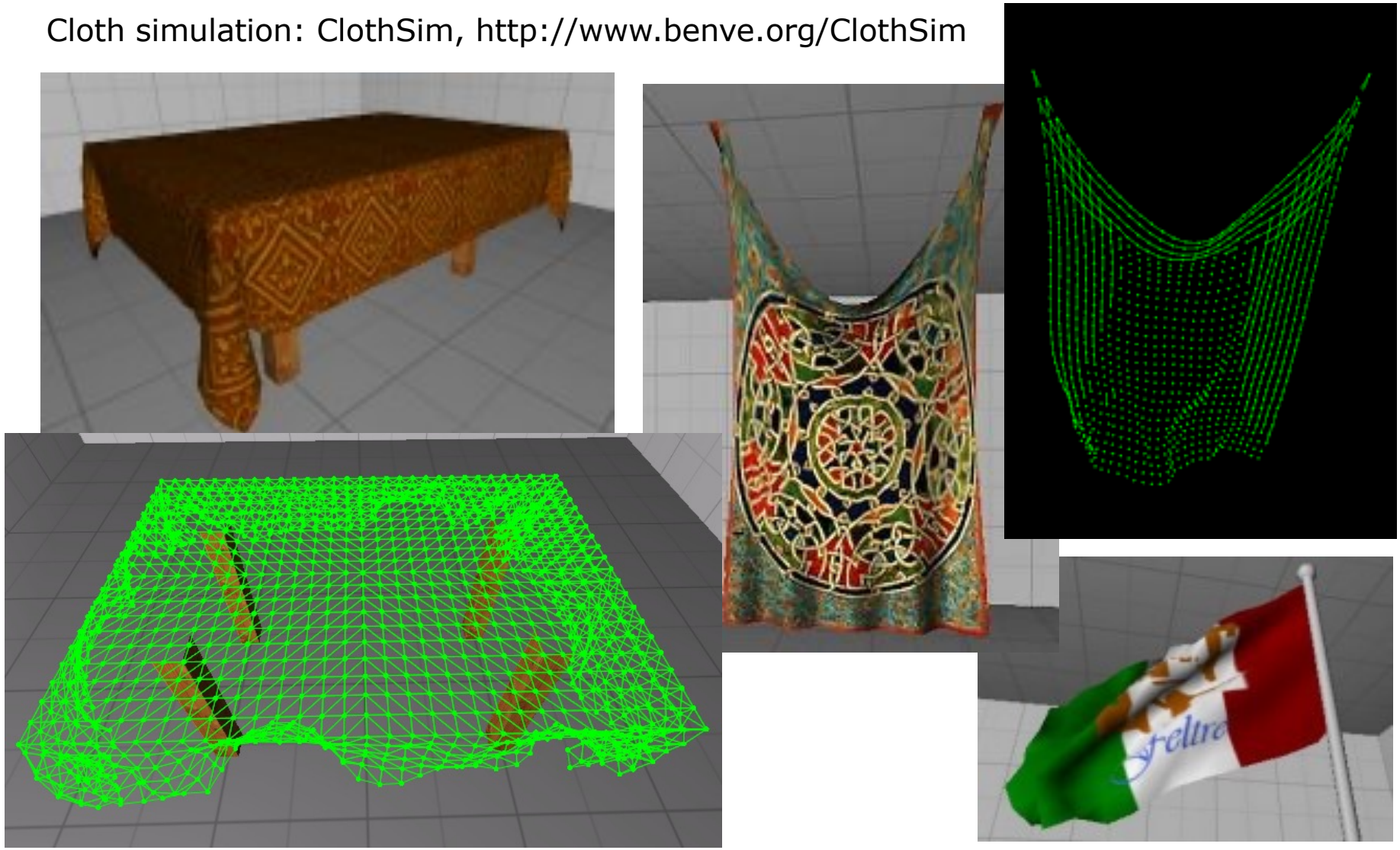
b) Shear springs



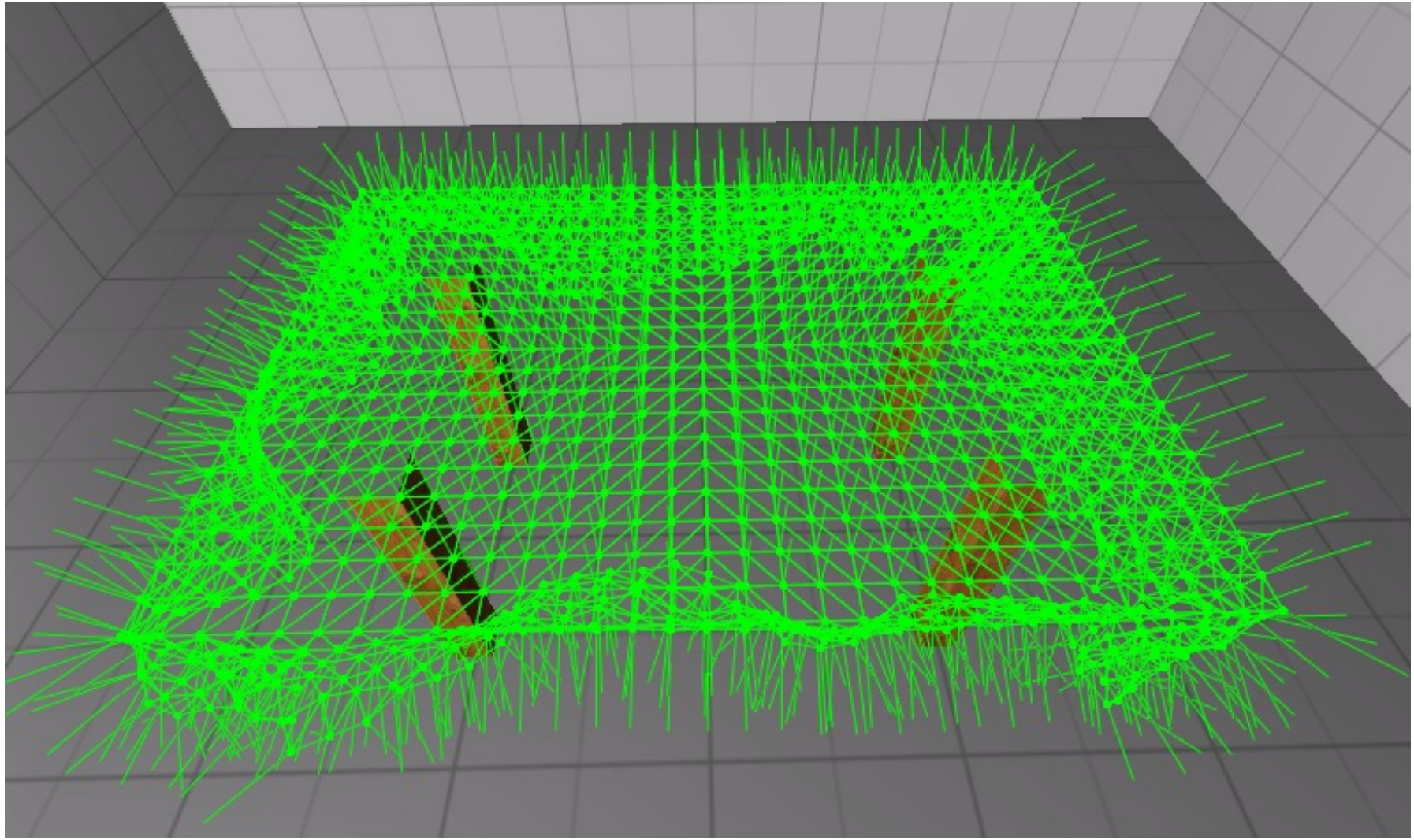
c) Bend springs
(not all connections shown)

Particle based modeling

Cloth simulation: ClothSim, http://www.benve.org/ClothSim



Particle based modeling - ClothSim

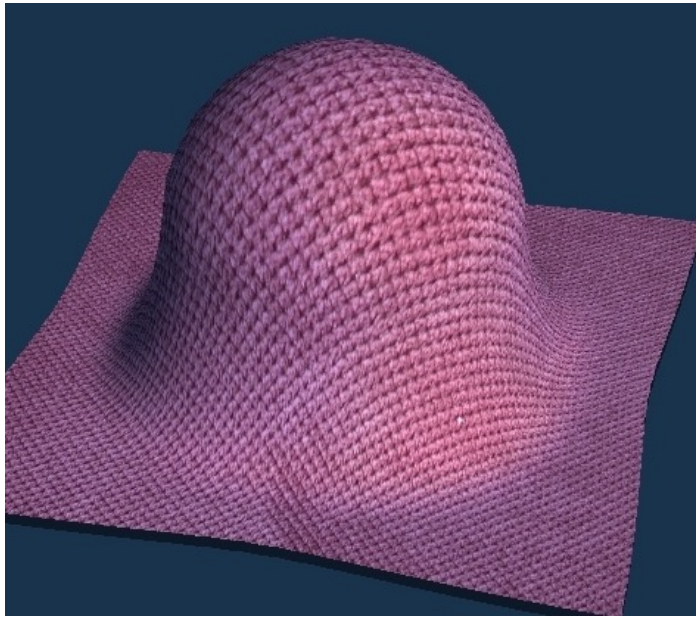


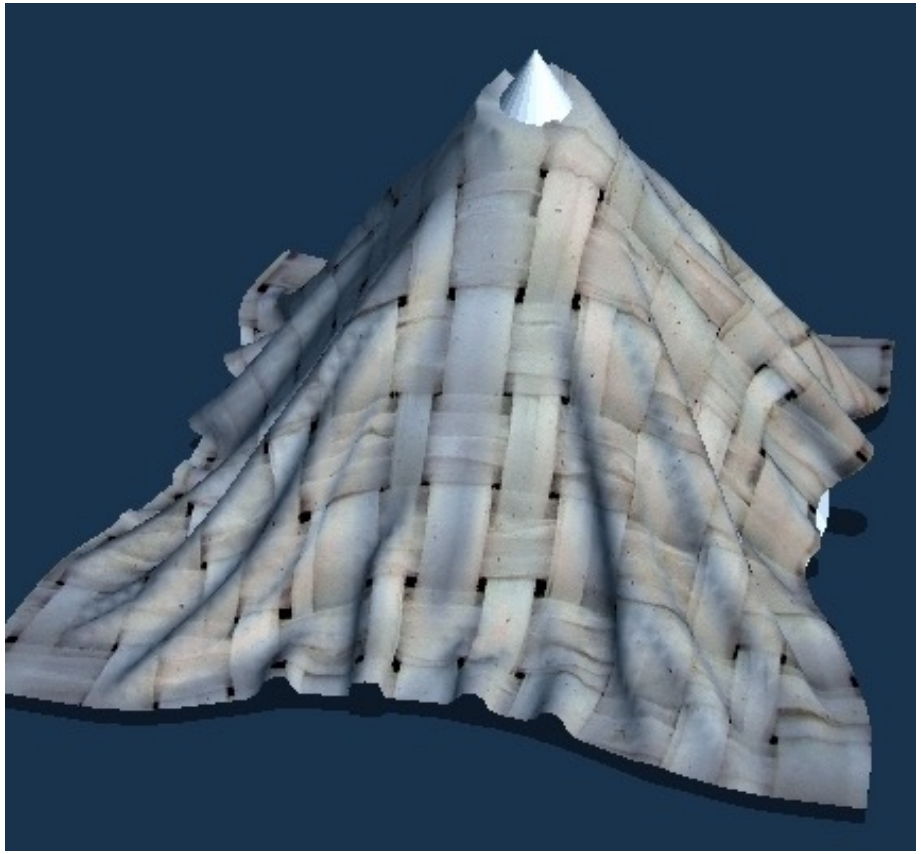
Particle based modeling



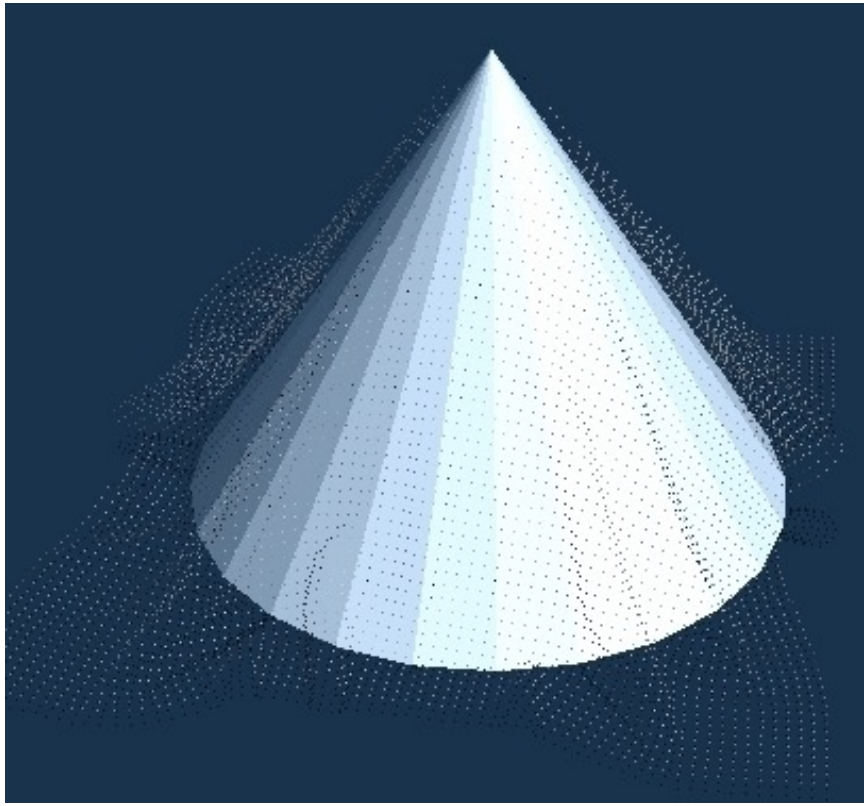


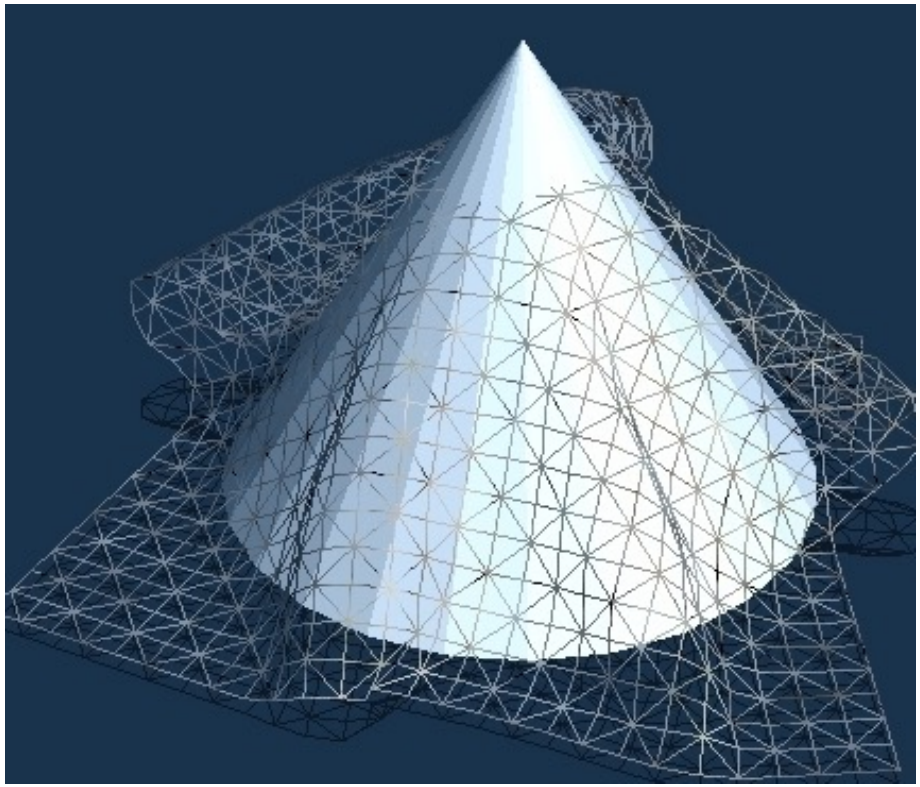
Particle based modeling



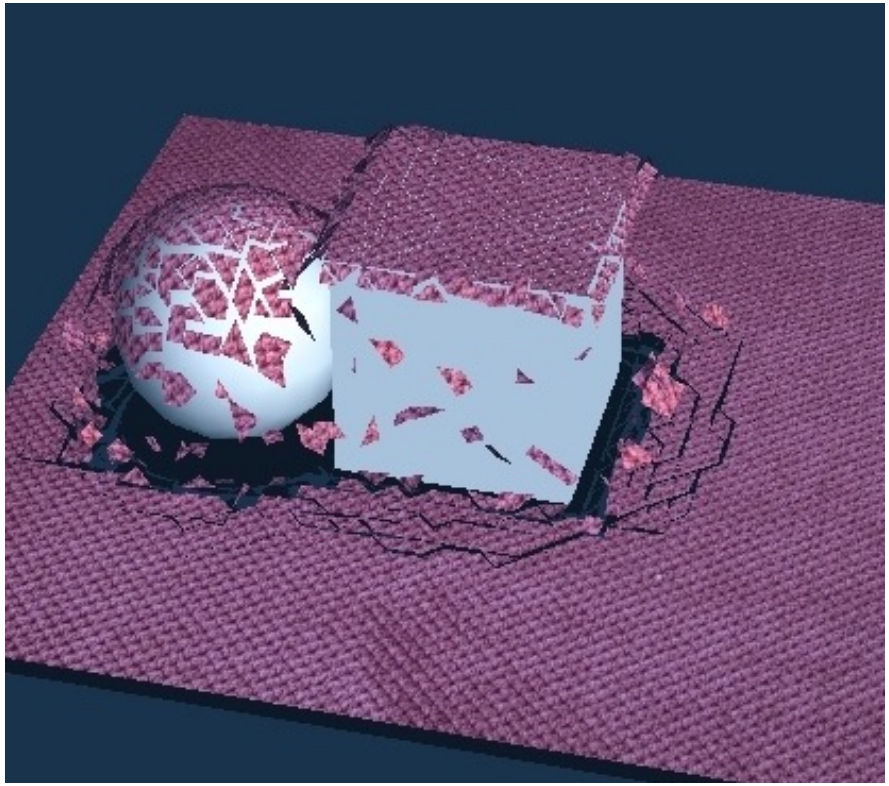


Particle based modeling





Particle based modeling



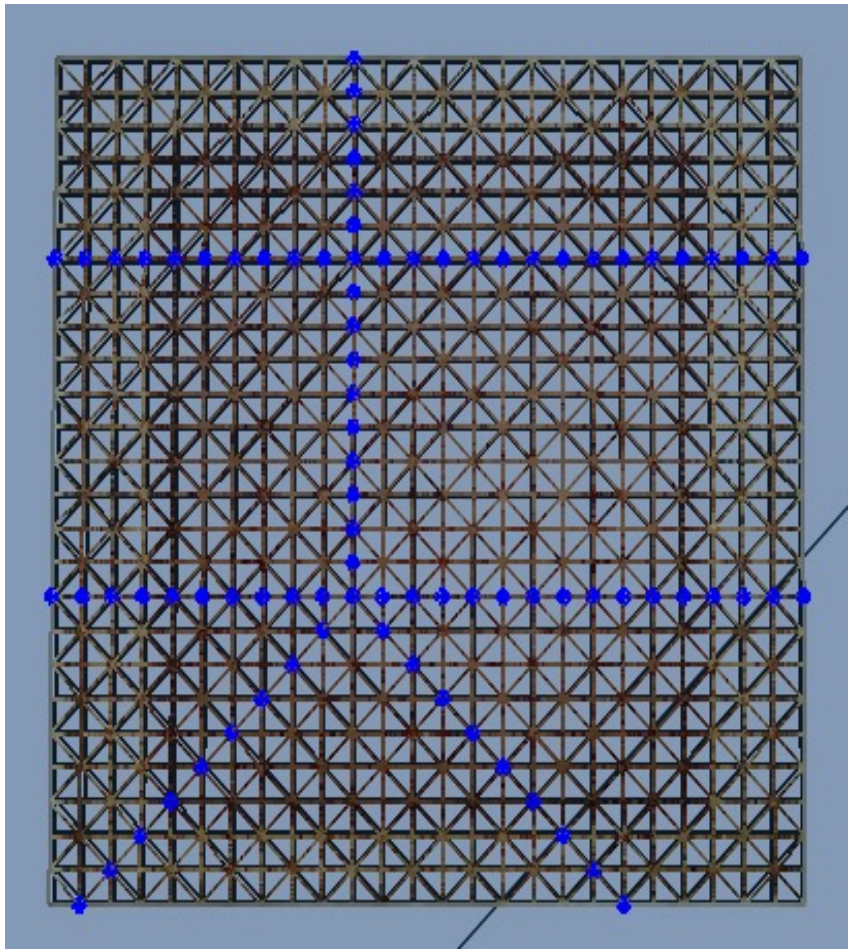


Particle based modeling





Particle based modeling - cutting

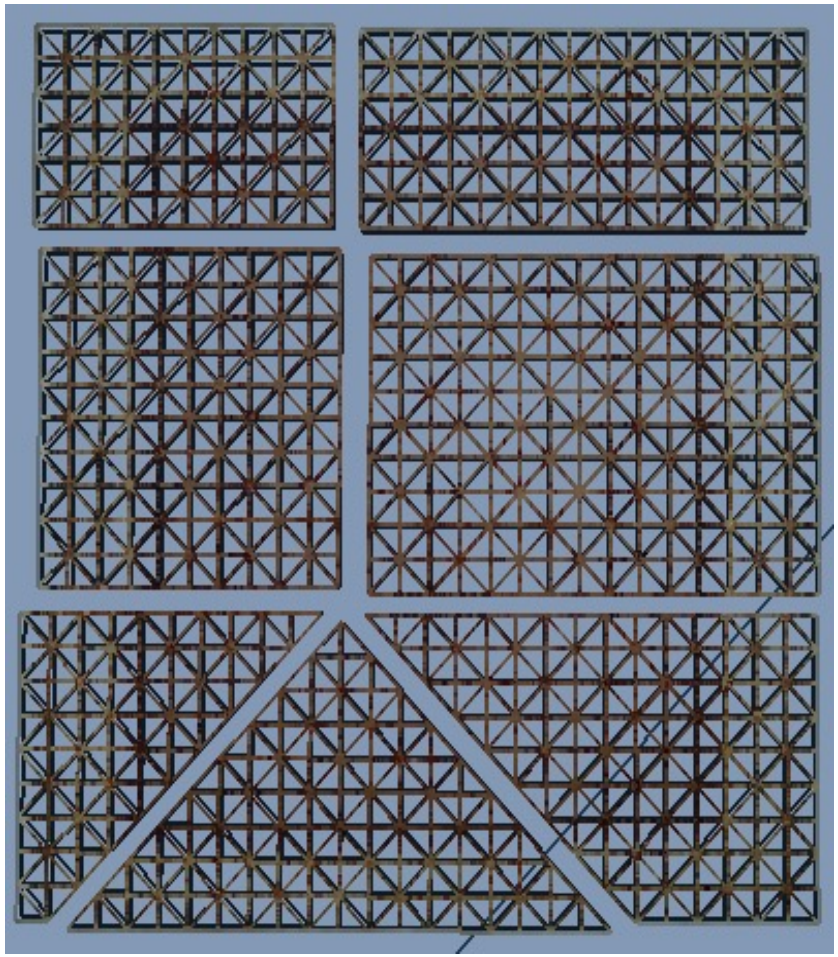


Selection of the cutting lines. Wireframe visualization.



Selection of the cutting lines. Texture mapping visualization.

Particle based modeling - cutting



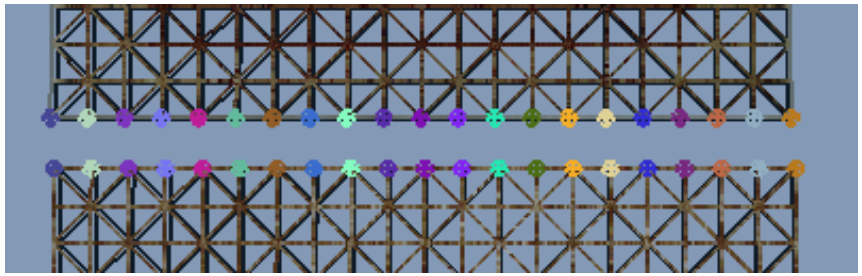
Cutting operation. W*ireframe visualization*.



Cutting operation. Texture mapping visualization.

Particle based modeling - sewing

Equal steps:



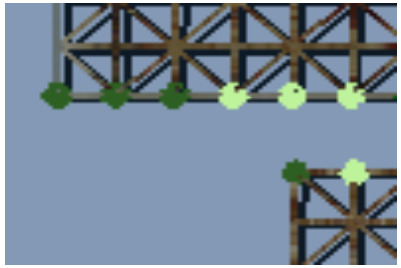
Preview of the cutting operation. W*ireframevisualization*.



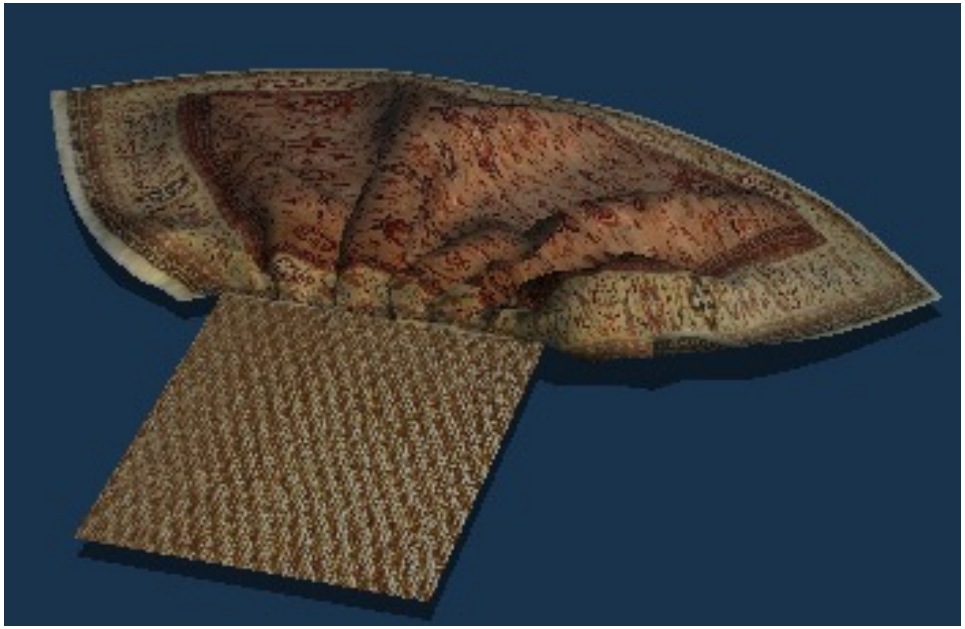
Sewing operation. Constant sewing step, egual one.

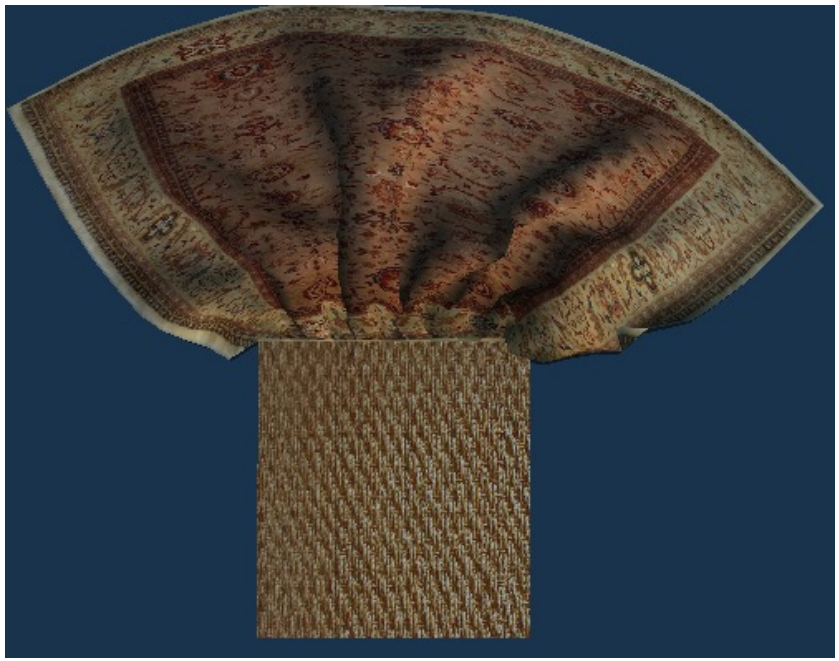
Particle based modeling - sewing

Sewing step equal three.



Preview of the cutting operation. Constant sewing step, egual three.

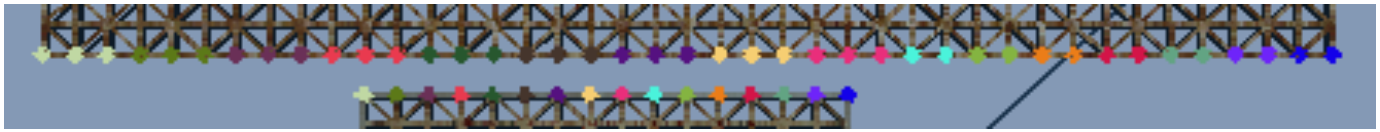




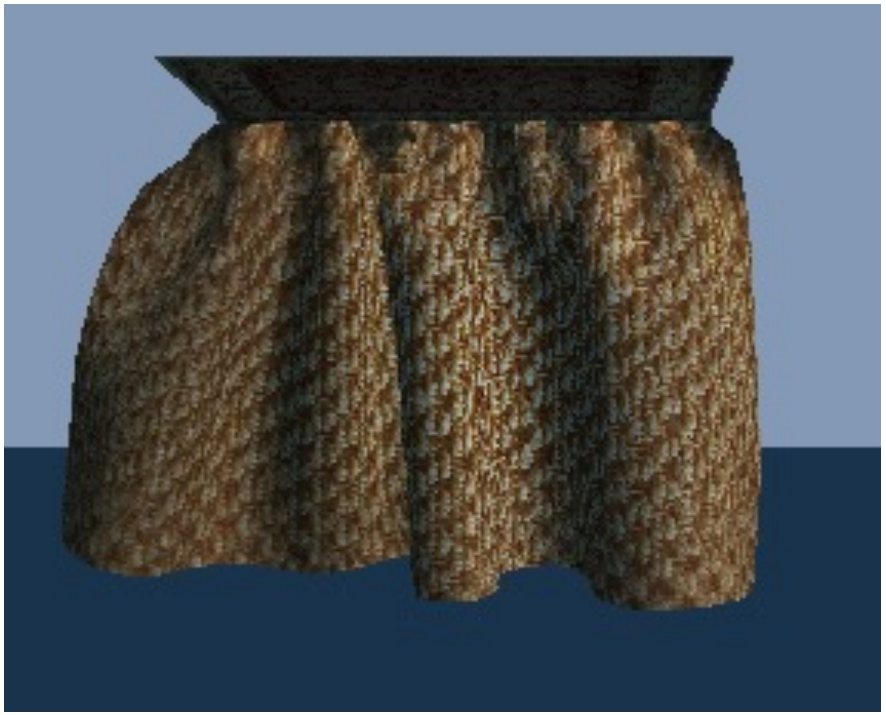
Sewing operation. Constant sewing step, equal three (Goga, 2009).

Particle based modeling - sewing

Variable sewing step.



Preview of the sewing operation. Variable sewing step, equal three and respectively two (Goga, 2009).



Sewing operation. Variable sewing step, equal three and respectively two (Goga, 2009).

Cloth on animated 3D objects



Baraff and Witkin, “Walking man and Dancer with long skirt”, SIGGRAPH 98 Conference.

Case studies

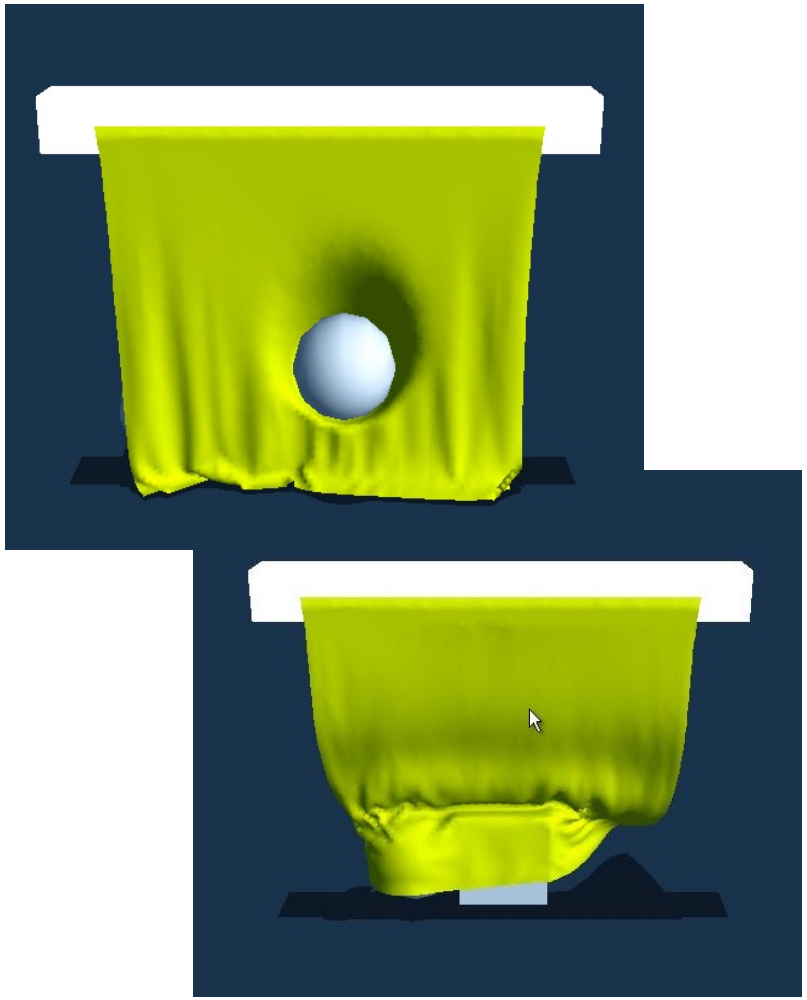
1. Distributed simulation and rendering
2. Parallel simulation
3. User interaction techniques

Use case 1 – Separated simulation and rendering

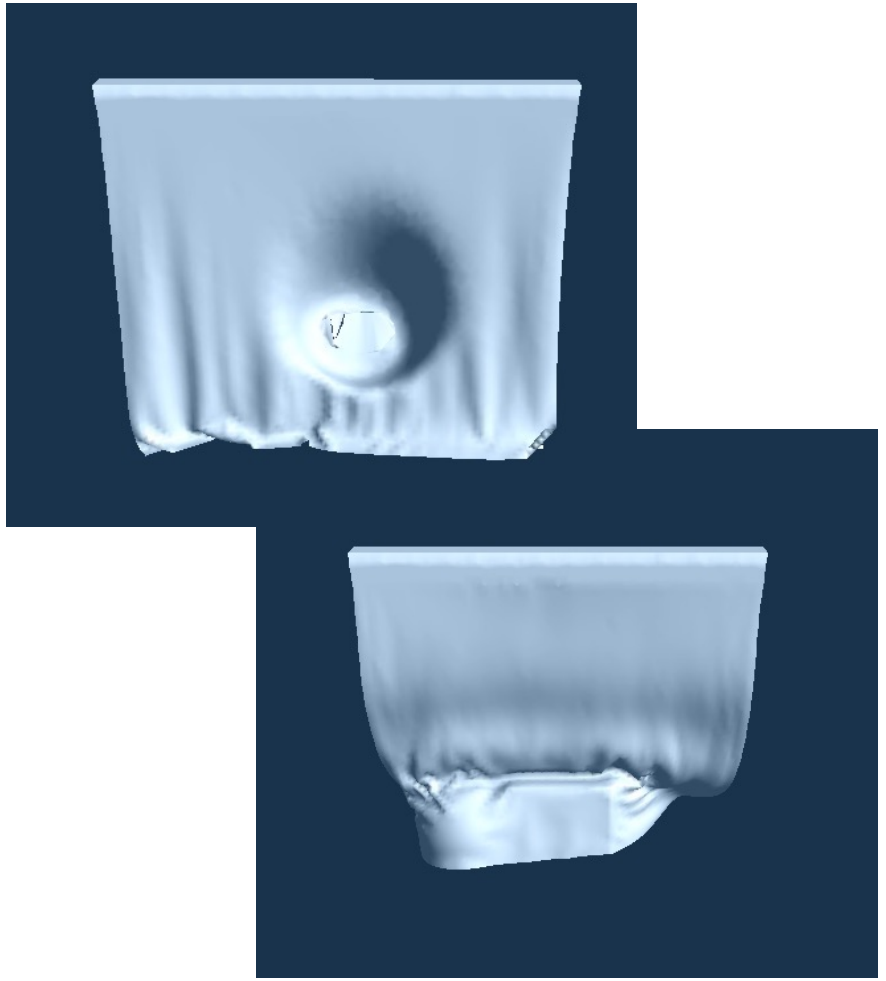


Use case 1 – Separated simulation and rendering

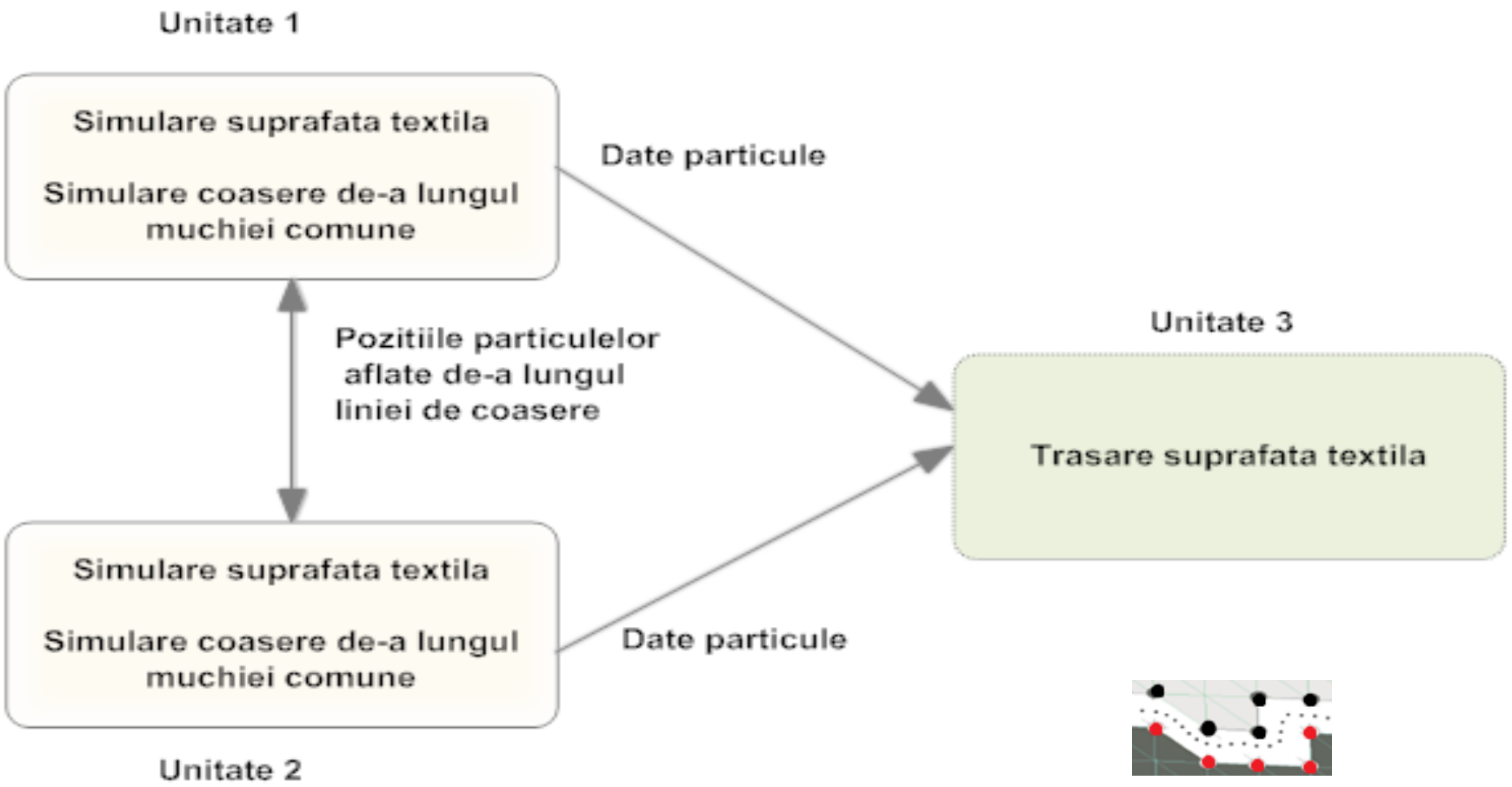
Textile surface on the simulation unit



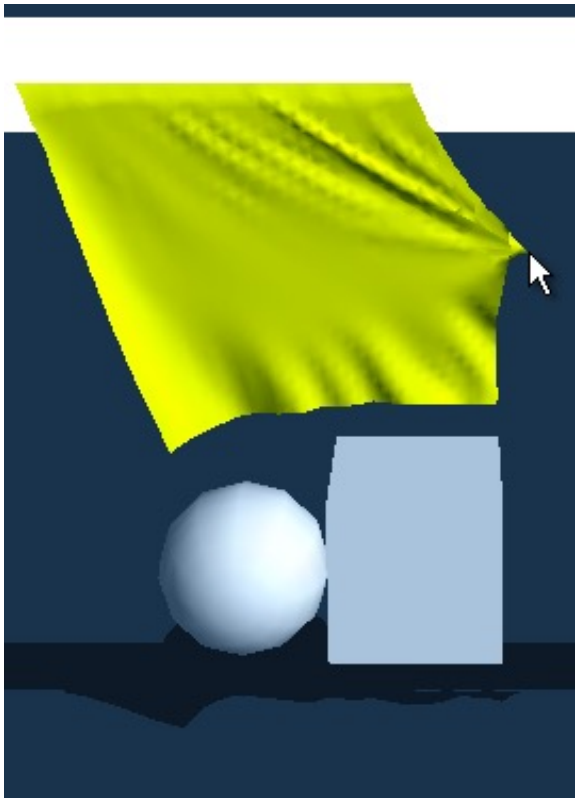
Textile surface on the rendering unit



Use case 2 – Parallel simulation



Use case 2 – Parallel simulation



Simulation unit 1

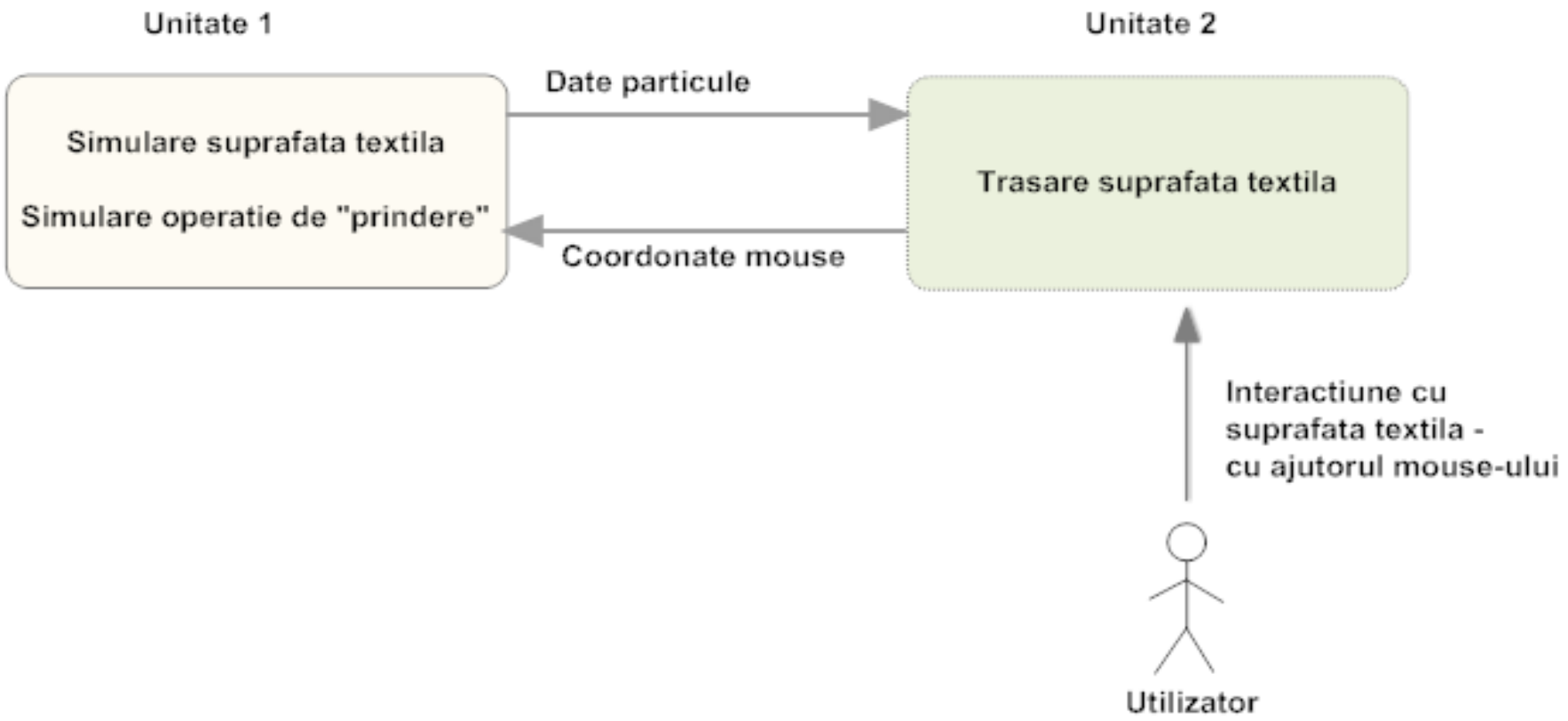


Simulation unit 2

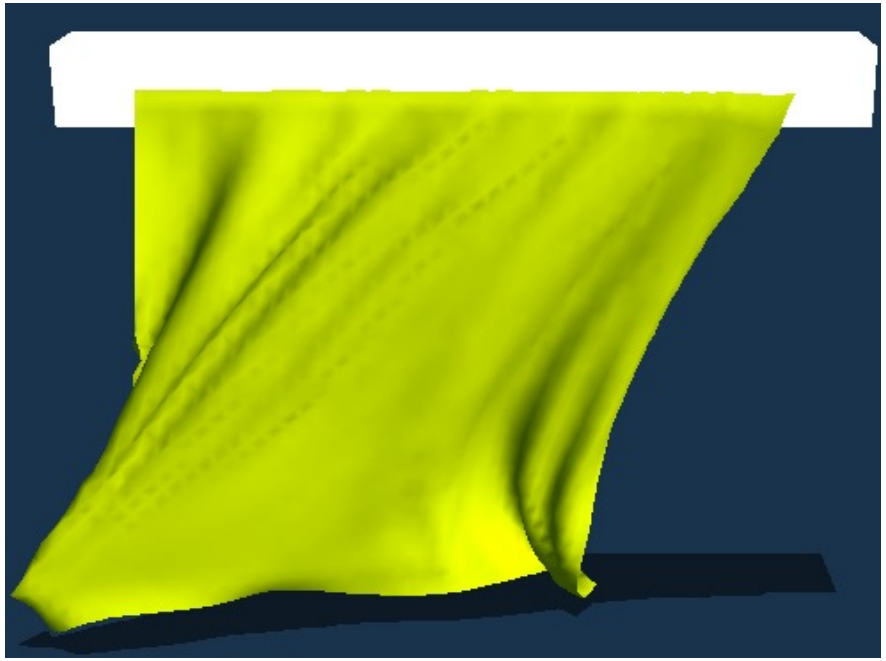


Rendering unit

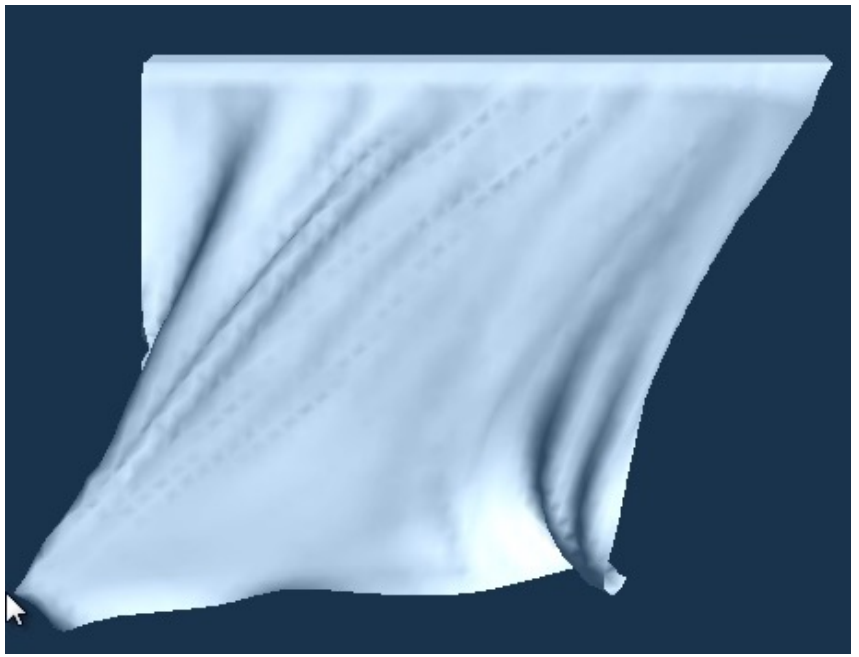
Use case 3 – User interaction



Use case 3 – User interaction



Simulation unit



User interaction at rendering unit

Questions and problems

1. Exemplify and explain the impact of the graphics processing objectives on the type and content of the graphics model.
2. Exemplify and explain the impact of the graphics model on the graphics processing pipeline resources (1) data structure; (2) processing algorithm; (3) design of software packages and hardware modules; (4) final appearance of the object; (5) techniques for editing the shape of the object.
3. Exemplify graphics model for a tetrahedron (regular triangular pyramid) in order to support the photorealistic presentation by BSP trees, shadows, and textures.
4. Why the triangle based polygonal model for the 3D object's surface is the most used?
5. Define and explain a fractal.
6. What is the fractal distance?

Questions and problems

7. What is the L-System grammar? Give examples of the grammar productions.
8. What features can be obtained by using the time based rules?
9. Explain and exemplify the particle based models.
10. How describes the mass-spring model the weight and the dimension of a piece of textile?
11. What issues arise the cutting and sewing operations on a mass spring model of a textile surface?